



Game Design Document

Exsanguination

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Game Design Document

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1 Game Concept

1.1 Setting

In *Exsanguination*, you play as a former vampire hunter who has recently turned into a vampire. Being a vampire is an improvement, but you must now use your new abilities to fight hordes of vampires who seek to destroy you. Explore a dark, gloomy world where speed and brutal action are key to survival in this fast-paced, first-person melee combat game. You'll use gadgets from your previous life to kill vampires swiftly—or be killed faster.

1.2 Genre

The genre is action, fast-paced first-person melee combat.

1.3 Platform

The game is built for PC with Windows as operating system, with 1920 x 1080p resolution as default.

1.4 Target audience

The target customers are primarily young adult male gamers between the ages of 18 and 35 (Pew Research Center, 2015). This demographic typically plays action-adventure games (Play Today, 2023).

2 Gameplay

2.1 Core Game Play

Intense, fast-paced first-person melee combat where survival hinges on quick reflexes and deadly precision. Cut down enemies with swords and melee weapons, use a stake to slow and finish off your foes, and use grappling hook to reach new heights. In this "go fast or die fast" world, parry incoming projectiles or attacks, dash and slide to dodge danger as you fight for your life.

2.2 Game Flow Chart

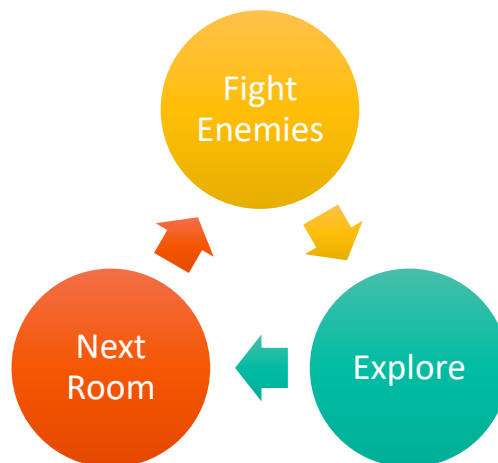


Figure 1: Game flow chart.

2.3 Gameplay Elements

2.3.1 Level layout

The game's level consists of 7, non-randomized rooms including the starting room where player gets to learn the mechanics of the game: basic movement, slide, jump and dash. After the starting room player's have their first combat encounter. The total layout of the level is as follows: starting room, combat room, hallway one, combat room two, hallway two, combat room three and the

boss room, totalling 7 rooms (Figure 2).

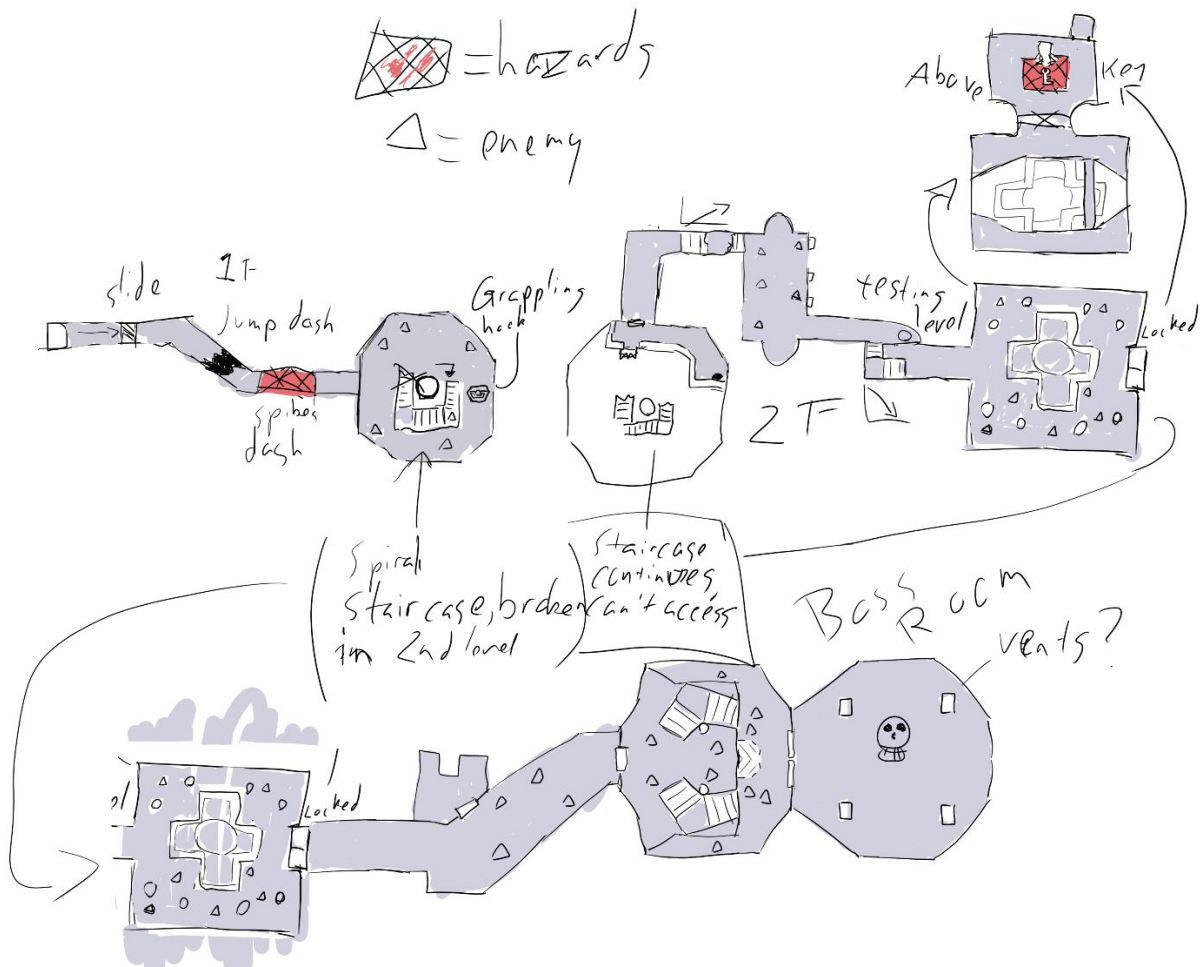


Figure 2: Level layout.

2.3.2 Gameplay run

Each run starts with similar set up: full health and starter sword with Dash and Slide unlocked. Player can find two crucial items of vampire hunter's arsenal: Stake and Grappling hook. Stake allows player to finish an enemy with 50% or less health while grappling hook allows player to gain new heights and aid in combat with fast movement. Before the boss room player receives health drops to make sure they are full HP. If the player dies anywhere in the level before the boss, they must restart over from the beginning, but if the player reaches the boss and dies in the boss fight, they can restart from the boss fight. Since players should have full HP in the start of the boss fight, whenever they restart the fight, they start with full HP. If the player has found second sword in the

level, it is acknowledged in the code, allowing players to have it in the start of the boss fight if they die.

2.3.3 Game-run stats

For each gameplay there are different stats that are stored in the data for the player, and they are displayed if they finish the boss. The stats include level completion time that resets if player dies in level, boss completion time that resets if player dies in boss fight, total time: combination of boss and level completion times and amount of player deaths.

2.3.4 Inventory and Resources

Pickable items in the level include the second weapon, silver bombs, health potions. Player can carry up to three silver bombs found within the level. Weapons and silver bombs are stored in equipment, and potions are used immediately on pick up.

2.3.4.1 Health potions

Picking up health potion restores 50% of max health instantly, cannot be held in inventory.

3 Game Mechanics

3.1 Player

3.1.1 Key inputs (controller)

Action	Key	Effect	Technical Data
Forward	W (Left stick)	Move forward.	Move speed = 7.
Left	A (Left stick)	Move left.	Move speed = 7.
Backwards	S (Left stick)	Move backward.	Move speed = 7.
Right	D (Left stick)	Move right.	Move speed = 7.
Jump	Space (A)	Jump up.	Move speed = 7.
Dash	Left-Shift (X)	Player moves rapidly to the direction they are pressing on WASD/Left stick.	Dash speed is 40. Dash time is 0.15 seconds. Dash cooldown is 2 seconds.

Slide	Left-Control (B)	Player gains speed and goes lower.	Slide time 0.5 seconds. Player speed doubled while sliding. Can slide under certain obstacles. Can slide on walls and does not lose height for the duration of slide.
Grapple	Q (Left shoulder)	Grappling hook can be used on certain points in the level to reach new heights.	Hook attaches to: hook points and enemies, player is pulled towards them. Grappling hook cooldown is 1 second.
Stake	E (Left trigger)	Stake can be thrown into enemies to slow them down and to finish them.	Enemy speed -50% when stuck on stake. If the enemy is below 25% health and has stake stuck on them, by pressing Use near them will trigger finisher -> enemy dies and player gains +50% health. Stake returns automatically to player after 5 seconds if not stuck on

			an enemy. If stuck on enemy and they are not at finisher threshold and player presses Use, the stick is returned to player without finisher.
Use	F (Y)	Allows the player to interact with level interactives such as levers and doors. Also allows for item pickups stake and weapons. The Interaction is also used for finisher.	
Attack	Left Mouse Button (Right trigger)	Attack with your currently equipped weapon.	With starter sword the player can combo with precisely timed clicks, increasing the damage up to 30 with the third attack.
Block	Right Mouse Button (Right shoulder)	With starter sword only: When timed correctly, it can be used to block melee attacks.	If blocked perfectly, stuns the enemy for 2 seconds and triggers the third attack and if it

			hits the enemy it deals third attack damage.
Special item	G (D-pad up)	Throws silver bomb.	Explodes and deals AoE damage of 100.
Equip starter sword	1 (D-pad left)	Equips the starter weapon.	
Equip SlayMore	2 (D-pad right)	Equips SlayMore if the player has found it within the level.	

Table 3: Input system explained.

3.1.2 Melee weapons

Player has access to two different weapons with different mechanics. Starter sword can combo attacks that allows for faster damage and third attack with big damage. It also has possibility to block melee attacks which stuns the enemy and counterattacks quickly, which if hit deals third attack damage. Slaymore (deprecated) is slow and deals a lot of damage. Left mouse button is big swing in front and right click is aoe swing where player spins around.

3.1.3 Melee Weapon Details

Name	Damage	Attack speed (s)	Special

Starter Sword	15	1.66	Combo: if successfully done, the third attack deals triple damage (45).
Slaymore (deprecated)	40	2	Left click for big swing in front and right click for aoe swing, both deals 40 damage.

Table 1: Melee Weapons, their stats and special effects.

3.1.4 Melee weapon mechanics detail

Sword

Starting item. Medium speed.

Great Sword – Slaymore (deprecated)

Can be found. Extremely strong, but slow to use.

3.1.5 Special weapons

Name	Damage	Area	Amount in level	Special effect
Silver bomb	33 DPS, hits every 0.3 seconds and dealing	10 units	3 bombs	Create silver caltrops. Deals damage every 0.3 second. It

	total of 165 damage.			lasts for 5 seconds. Deals 10 damage per hit. Player does not take damage from caltrops.
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Table 2: Special weapon silver bomb, it's stats, info and special effect.

3.1.6 Special weapon mechanics detail

Alchemy bomb

The Silver Bomb deals 33 DPS, delivering a total of 165 damage over 5 seconds. Upon detonation, it creates a field of silver caltrops in a 10-unit radius, which inflict 10 damage every 0.3 seconds. The player is immune to damage from the caltrops, and up to three bombs can be carried at a time.

3.2 Enemies

Enemies have idle, see, chase and attack states. Ranged enemies might not start the chase state if they are within range.

3.2.1 Common melee enemies

3.2.1.1 Ghoul

Runs fast and claws fast, deals moderate damage.

Name	Health	Damage	Attack range	Attack cooldown	Movement speed	Details
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Ghoul	20	10	1.5	1.4	4	Undead, ground enemy. Strikes twice slowly.
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3.2.2 Common ranged enemies

3.2.2.1 Bat

Flying enemy. Avoids low ground and strikes at the player from afar. Highly annoying poke damage.

Name	Health	Damage	Attack range	Attack cooldown	Movement speed	Flying height	Details
Bat	10	10	7	3	2	3 units	Shoots sonic projectiles whilst flying.

3.2.3 Elite enemies

3.2.3.1 Vampire (elite mob)

Big and nasty, throws blood balls that can be reflected. If close uses bite attack that can heal them.

Name	Health	Damage	Attack range	Attack cooldown	Movement speed	Details
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Half blood	25	20	2	3	25	Heals half of the damage dealt to itself
Full blood	50	40	2	2.5	30	Heals the damage dealt to itself

3.2.4 Spawn mechanics

3.2.4.1 Entering a room

When player enters a trigger, they aggro that wave's enemies. In a room there can be multiple triggers that trigger specific enemies of the area. When enemies are triggered they start to chase the player until either is dead.

3.2.5 Boss

Health: 200.

At 10% health, boss will escape the room, avoiding death.

3.2.5.1 Boss Dash

When boss dashes away, they are invulnerable and can go through walls. Boss chooses random waypoint that is at least 5 units from player, then faces towards the waypoint and dashes towards it. When boss reaches the waypoint they start facing player and changes state.

3.2.5.2 Melee attacks

Damage: 80

Description: The boss dashes towards the player, they slash once towards the player, dealing 80 damage on a successful hit. If boss hits successfully the player, they dash away immediately. If boss's attack is blocked successfully, the boss is stunned for 2 seconds where the boss cannot attack or move. After the stun is over, boss dashes.

3.2.5.3 AOE attacks

Ground based AOE attacks linger for 10s after cast, marked with *.

3.2.5.3.1 Pirouette

Boss casts for 1.5 seconds, then throws daggers in front of them and spins 360 degrees quickly, which causes daggers to fly whole 360 rotation around the boss. Daggers are thrown every 0.1 seconds and for 1 second, 10 daggers in total. Daggers can be reflected and they deal damage to boss. Each dagger deals 20 damage.

3.2.5.3.2 Spike growth

After casting for 2 seconds, spawn spike trap under player on the floor. X and Z value come from player and Y is always the same (floor is on same level). After 10 seconds the spike disappears. Spikes deal 20 damage if hit. Works same way as spikes in the level.

3.2.5.3.3 Firewall

Boss casts firewall that starts from the boss's current location towards player's current location, going up to 10 units. This means it goes over the player's location. There is a smouldering line that spawns first in 0.5 seconds and after in 1.5 seconds the fire wall starts to emerge from boss and moves to player's last location fast speed taking 0.5 seconds. The wall stays up for 5 seconds then goes down in 0.5 seconds and disappears. Hitting the wall player takes 30 damage.

3.2.5.3.4 Hellfire

Boss casts for 1 second, then shoots homing projectile towards player. It does not destroy until it hits something, wall or pillar or the player. If it hits player, it deals 30 damage, cannot be reflected, if prompted it just damages the player.

3.2.5.4 Attack cycle

- Melee attack
 - If blocked
 - stun 2 seconds
 - dash away to random waypoint that is at least 5 units from player
 - If not
 - dash away to random waypoint that is at least 5 units from player
- Idle 5 seconds
 - dash away to random waypoint that is at least 5 units from player
- Special attack
 - After casting special attack
 - dash away to random waypoint that is at least 5 units from player
- Idle 5/3/1/0 seconds
 - dash away to random waypoint that is at least 5 units from player
- Melee attack
- Repeat

3.2.5.5 Environment

Room has 16 vents, where they are activated once the boss reaches 150, 100 and 50 health. At 150 health 3 vents are activated, at 100 health 4 more are activated and at 50 health 7 more are activated, totalling 14 active vents and 2 inactive vents. Vents release a cloudy gas that deals 34 damage per second for each second spent in the gas.

4 Player feedback systems

4.1 Hud elements

4.1.1 Health

Visual splatter on screen that increases the lower the health of the player.

4.2 In-game notifications

Press F to open door, read book etc.

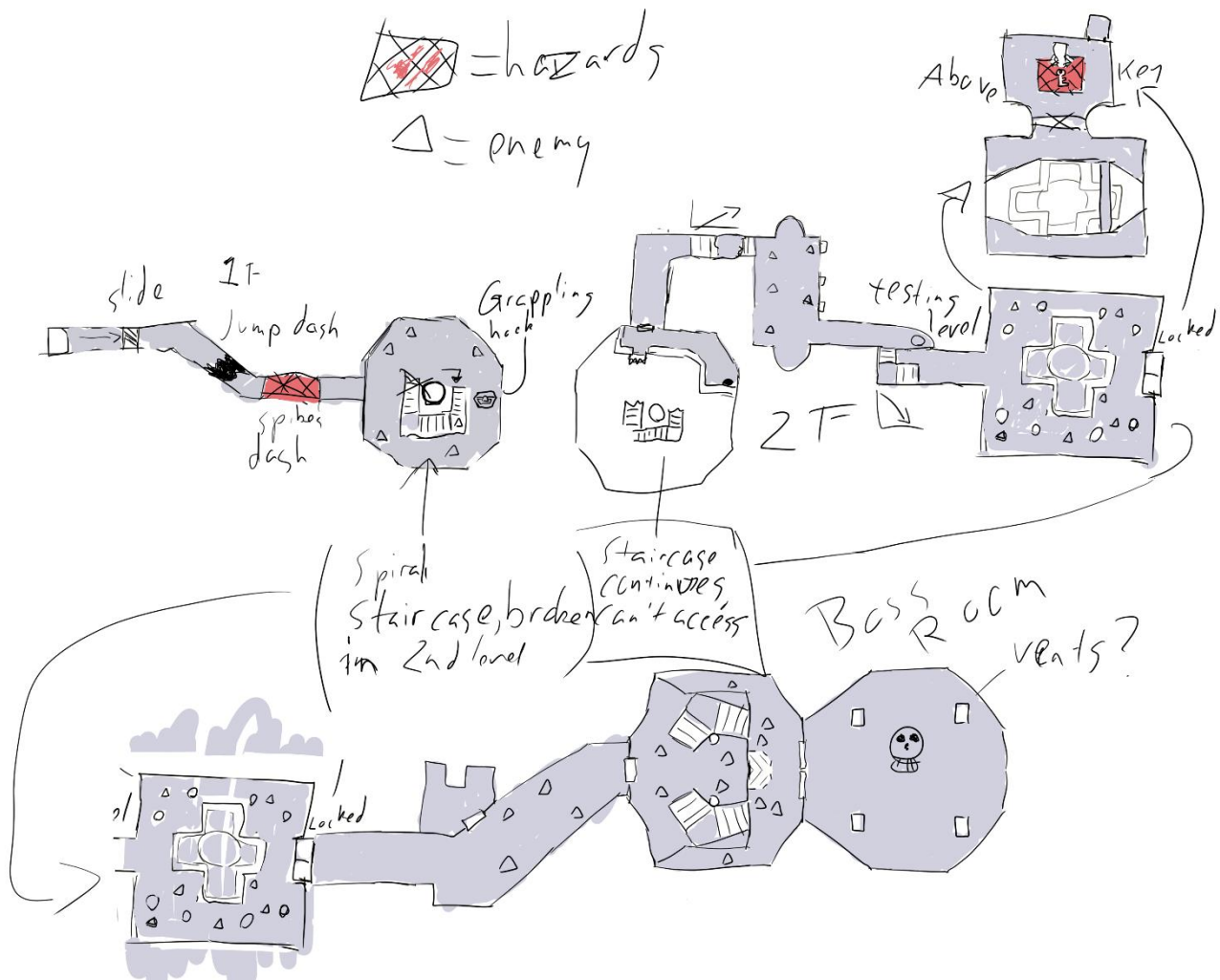
4.3 Visual cues

- glows when u see grappling hook spot.
- Finisher visual
- Low health visual cues
- Health pick up cues

5 Level

5.1 Layout and rooms

Picture:



5.2 Item spawns

The items spawned in the level are as follows: Stake in tutorial, at the end of it where you are taught how to finish an enemy, Grappling hook at the midpoint of stairway spiral room, Key to open a door to boss room and lastly one hidden health potion.

5.3 Spawns

5.3.1 Player spawn

Level consists of three different spawns: start, grappling hook tutorial room and grappling hook puzzle room. Falling into pits will reset you at the beginning whether you are in grappling hook tutorial room or puzzle room and dying in any parts of the level puts you right in the beginning. Falling in tutorial room will put you at the beginning and dying in boss room starts the boss fight over.

5.3.2 Enemy Spawn

Enemyspawns are as follows: 4 in downstairs of the level, 2 in the staircase, 2 at the grappling hook pickup, four in upper side of the staircase room, 4 in the lower section of main room, 4 in upper side and 3 in the key room. Key room enemies are all bats and one of the upper side of the staircase room enemies are a bat.

5.4 Traps and hazards

Tutorial has two pits, level has multiple and spikes are located under the key and on some walls during grapple hook puzzle.

References

Pew Research Center. (2015). *Gaming and gamers*. <https://www.pewresearch.org/inter-net/2015/12/15/gaming-and-gamers/>.

Play Today. (n.d.). *Gamer demographics: Stats and trends for 2023*. Play Today. <https://playtoday.co/blog/stats/gamer-demographics/>.

Appendices

Appendix 1. Title of the Appendix